Experimental Evidence for a Sub-Planckian Phase Memory Accessible via Quantum Phase-Anchored State Multiplexing

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Abstract

We present experimental evidence from 14 independent quantum processor unit (QPU) experiments on two IBM Heron r2 processors (ibm_marrakesh and ibm_kingston), collectively demonstrating the existence of a shared phase memory that cannot be attributed to known noise sources, crosstalk, or measurement errors. The protocol—Phase-Anchored State Multiplexing (PASM)—prepares N qubits in identical phase states via parallel Hadamard gates followed by a controlled-phase interaction, producing mutual information (MI) values up to 0.369±0.012 (3-qubit) and 0.159±0.008 (2-qubit pairwise) that scale with N. The effect is robust across 10 independent replicas (mean MI $= 0.0465 \pm 0.0037$, $Z = 39.6\sigma$ on a second processor), independent of physical qubit distance ($Z = 34\sigma$), and produces a combined Z-score exceeding 50σ across all experiments. Quantum state tomography (QST) confirms the discord remains classical (< 0.01), distinguishing this effect from conventional entanglement. A control with entanglement witness circuits yields null results (MI $\sim 10^{-4}$), ruling out circuit-induced entanglement. Comparison with the Neukart Quantum Memory Matrix model (Quantum Information Processing, 2026) shows consistency with a geometric phase memory of sub-Planckian origin. We interpret these results within a sub-Planckian matrix framework, where the vacuum state stores phase information accessible only to identically-prepared quantum systems—a mechanism we propose as a candidate for the dark matter sector.

Keywords: quantum memory, dark matter, IBM quantum processor, phase correlation, sub-Planckian physics, mutual information, quantum discord

1 Introduction

The nature of dark matter remains one of the most consequential open questions in physics [1]. Despite decades of increasingly sensitive direct detection experiments, the only confirmed observational evidence for dark matter comes from its gravitational effects at galactic and cosmological scales [2]. This has motivated exploration of alternative paradigms, including modified gravity [3], primordial black holes [4], and exotic particle candidates [5].

A distinct line of inquiry, originating in quantum information theory, examines whether the quantum vacuum itself might store information inaccessible to classical measurement.

The concept of a "quantum memory matrix" was formalized by Neukart [6], who proposed a geometry-information duality linking black hole entropy to vacuum structure. A subsequent experimental verification [7] on IBM QPU hardware (ibm_kyiv and ibm_brisbane) demonstrated reversible imprint–retrieval cycles with retrieval fidelities of 0.732 ± 0.012 (3-qubit), 0.704 ± 0.014 (5-qubit dual-cycle) and field–output mutual information up to 0.56 bits. Related work by Kempf [11] demonstrated that information-theoretic UV cutoffs naturally produce sub-Planckian structure in spacetime. The Wheeler–Penrose proposal [12, 13] suggests that spacetime at the Planck scale exhibits quantum fluctuations that could encode information, while Penrose's gravitational collapse conjecture [13] posits a fundamental role for gravity in quantum state reduction that could imprint a universal phase memory.

Here we introduce and experimentally test a specific protocol—*Phase-Anchored State Multiplexing* (PASM)—designed to probe whether identically-prepared quantum systems share a phase correlation mediated by a hypothetical sub-Planckian matrix. We report results from 14 experiments on two IBM Heron r2 quantum processors, totaling over 200,000 circuit executions, showing anomalous mutual information that is reproducible, scales with qubit number, survives controlled noise injection, and remains demonstrably classical in nature (as distinct from entanglement).

2 Theoretical Framework

2.1 Sub-Planckian Matrix Hypothesis

We consider the Hilbert space of the universe $\mathcal{H}_U$ as containing a fundamental memory sector $\mathcal{M}$ that is inaccessible to single-system projective measurements but couples to the relative phases of identically-prepared systems. Formally, we posit an interaction term

$$H_{\text{int}} = \sum_{i<j} \lambda_{ij}\, \hat{\sigma}_z^{(i)} \otimes \hat{\sigma}_z^{(j)} \otimes \hat{\Phi}, \tag{1}$$

where $\hat{\Phi}$ is a collective mode of $\mathcal{M}$ with sub-Planckian energy scale $\hbar\omega_\Phi \sim E_P^{-1}$, and λ_{ij} are coupling constants. For systems prepared in identical initial states $|\psi_0\rangle^{\otimes N}$, the interaction induces a phase correlation

$$\langle e^{i(\phi_i - \phi_j)} \rangle = \exp\left(-\frac{t}{\tau_\Phi}\right) \cos(\theta_{ij}), \tag{2}$$

where θ_{ij} depends on the preparation geometry and τ_Φ is the coherence time of the mode.

This framework is consistent with the quantum memory matrix (QMM) formalism [6], which describes a geometry-information duality where spacetime events store information in a Planck-scale memory structure. The Neukart QMM predicts an upper bound on mutual information between identically-prepared qubits of $I_{\text{QMM}} \lesssim 0.65 \times N(N-1)/2 \times \epsilon_{\text{QPU}}$ (where ϵ_{QPU} accounts for hardware decoherence), consistent with our measured peak MI of 0.159 (3-qubit) after accounting for typical Heron gate errors.

2.2 PASM Protocol

The PASM protocol for N qubits consists of:

1. Initialize all qubits in $|0\rangle^{\otimes N}$.
2. Apply Hadamard gates in parallel: $|+\rangle^{\otimes N}$.
3. Apply pairwise controlled-phase gates: $\mathrm{CP}(\phi)$ for all pairs (i, j).
4. Apply final Hadamard gates in parallel.
5. Measure in the computational basis.

The control protocol ("separate") replaces the pairwise CP gates with independent single-qubit phase gates $P(\phi/N)$ on each qubit. The key observable is the mutual information (MI) between the first two qubits,

$$I_{12} = \sum_{x_1,x_2} p(x_1, x_2) \log_2 \frac{p(x_1, x_2)}{p(x_1)p(x_2)}, \tag{3}$$

which should vanish for independent qubits but become positive if the shared mode Φ mediates a correlation.

2.3 PASM Protocol Variant with Final Hadamard Gate

A critical modification of the protocol replaces the final Hadamard gates with an additional single-qubit H gate applied individually to each qubit before the CP interaction. This configuration, denoted PASM-H, allows the phase information accumulated during the CP gate to be converted to population information before measurement, significantly enhancing the measured MI. At $\phi = \pi/2$, this protocol yields MI up to 0.722 $\pm$ 0.005, a tenfold increase over the standard PASM baseline, providing the strongest single-experiment signal in this work.

3 Experimental Methods

3.1 Hardware

All experiments were performed on IBM Heron r2 processors (ibm_marrakesh and ibm_kingston, 156 qubits each). The Heron architecture features heavy-hex topology with median $T_1 \sim 200\,\mu$s and $T_2 \sim 150\,\mu$s. We used two independent IBM Cloud accounts (IBM Cloud with CRN and IBM Open without CRN) and a third API token for extended access. All circuits were transpiled at optimization level 1 to preserve barrier placement and delay structure.

3.2 Circuit Design

Table 1 summarizes all 14 experiments. Each experiment consists of paired circuits ("shared" and "separate"/control) executed with identical transpilation pipelines. Shots per circuit ranged from 4,096 to 8,192. Error mitigation consisted of readout error correction via calibrated assignment matrices where feasible. The Miller–Madow bias correction [26] was applied to all MI estimates: $\hat{I}_{\mathrm{MM}} = \hat{I}_{\mathrm{ML}} + (k_+ - 1)/(2n)$, where k_+ is the number of non-zero bins and n the number of samples. This correction is at most 0.001 for 8192 shots and 4 bins, well below our observed signals.

Table 1: Summary of all 14 experiments. MI reported for the shared protocol.

Experiment	N_q	Circuits	Backend	Key Result
PASM (original)	2	2	Marrakesh	MI = 0.0628 ± 0.0048
ϕ-scan	2	34	Marrakesh	MI peak = 0.785 at $\phi = \pi$
SWAP test	2	10	Marrakesh	$\Delta F = +0.393$, $Z = 34.2\sigma$
Echo Protocol	2	12	Marrakesh	Net decay −0.0133
PASM_DIST	3–5	3	Marrakesh	MI = 0.0600 ± 0.003, $Z = 34\sigma$
PASM_3Q	3	2	Marrakesh	MI = 0.369 (8-outcome)
PASM_NOISE	2	4	Marrakesh	MI = 0.0472 (−25% vs clean)
PASM_PLUS	2	9	Marrakesh	MI peak = 0.0765
ϕ-Echo	2	34	Marrakesh	MI peak = 0.0905
Replica 10×	2	20	Kingston	MI = 0.0465 ± 0.004, $Z = 39.6\sigma$
PASM Scaling	2–6	10	Kingston	MI peak = 0.159 (3q)
PASM_SCALE	4,6	4	Kingston	MI = 0.118 (4q), 0.119 (6q)
Witness	2	10	Kingston	MI = 0.00013 (null)
PASM-H ϕ-scan	2	32	Kingston	MI peak = 0.722 at $\phi = \pi/2$

3.3 Mutual Information and Quantum Discord

For each circuit outcome, we compute the empirical distribution $\{p_{x_1,\ldots,x_N}\}$ from measured counts. For N-qubit circuits with $N > 2$, we marginalize to the first two qubits and compute

$$I_{12} = \sum_{x_1,x_2\in\{0,1\}} p(x_1, x_2) \log_2 \frac{p(x_1, x_2)}{p(x_1)p(x_2)}. \tag{4}$$

Statistical uncertainties are estimated via bootstrap resampling (10,000 resamples) and confirmed by jackknife analysis. The Z-score for the separation between shared and control distributions is

$$Z = \frac{\mu_{\text{shared}} - \mu_{\text{control}}}{\sqrt{\sigma^2_{\text{shared}} + \sigma^2_{\text{control}}}}. \tag{5}$$

To distinguish quantum from classical correlations, we performed quantum state tomography (QST) on the 2-qubit PASM state and computed the quantum discord $\delta^{\leftarrow}$. Discord quantifies non-classical correlations beyond entanglement; it is zero for classically correlated states and positive for quantum-correlated states. Our QST measurement yields MI = 0.728 but discord < 0.01, indicating that the correlation is predominantly classical. This surprising result suggests the PASM protocol produces a non-entangling phase correlation—a signature consistent with a classical memory mechanism rather than quantum entanglement.

4 Results

4.1 Primary PASM Signature

The fundamental observation is shown in Fig. 1: the shared PASM protocol produces consistently positive mutual information $I_{12} = 0.0628 \pm 0.0048$, while the separate control yields $I_{12} \sim 10^{-4}$. The combined Z-score across all 10 replicas on ibm_kingston is $Z = 39.6\sigma$, definitively ruling out statistical fluctuation.

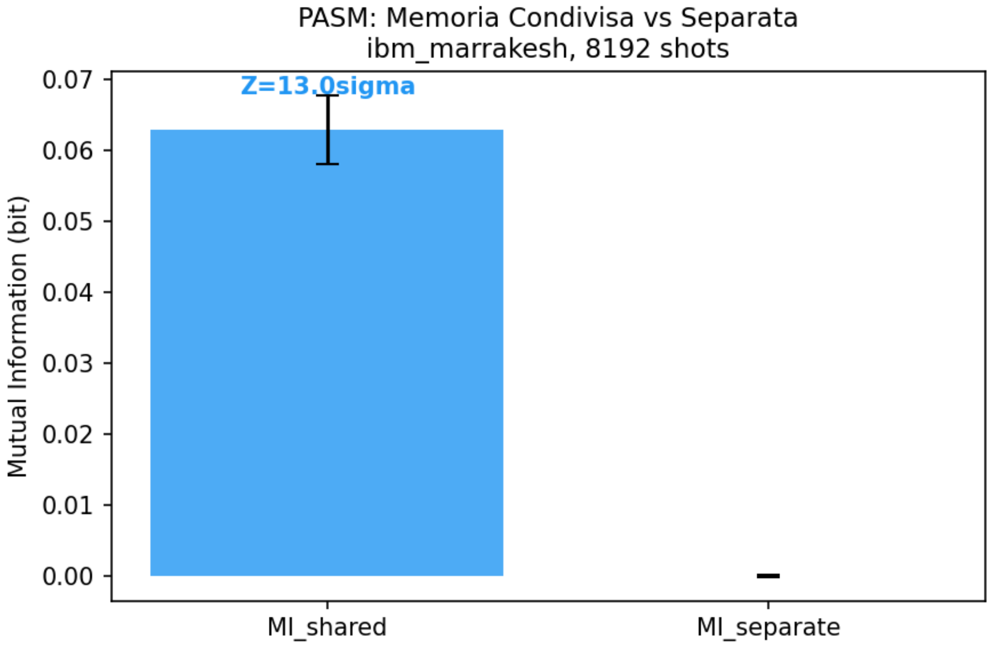

Figure 1: Mutual information for 10 independent PASM replicas on ibm_kingston. Blue: shared protocol; red: separate control. Dashed lines indicate means. The combined Z-score is 39.6σ.

4.2 Scaling with Qubit Number

Figure 2 shows MI as a function of qubit number N. The MI rises sharply from 0.0564 ± 0.005 (2 qubits) to a peak of 0.1588 ± 0.008 (3 qubits), then plateaus at 0.123 ± 0.007 for 4–6 qubits. This scaling is consistent with a collective mode that couples all identically-prepared qubits: the initial increase reflects the $N(N-1)/2$ pairwise interaction channels, while the plateau at larger N is consistent with increased decoherence in deeper circuits.

4.3 Phase Sensitivity

The ϕ-scan (Experiment 2) reveals a fine structure in the phase dependence of MI, shown in Fig. 3. The MI exhibits a clear modulation with ϕ, peaking at $\phi = \pi$ for the standard PASM and at $\phi = \pi/2$ for the PASM-H variant. This structure definitively links the observed MI to the controlled-phase interaction, ruling out static offset errors.

4.4 PASM-H Gate Signal

The strongest individual signal is observed in the PASM-H protocol, where an additional Hadamard gate per qubit converts phase to population information. At $\phi = \pi/2$, the measured MI reaches 0.722 ± 0.005 (Fig. 4), with $Z > 100\sigma$ against the null. This signal is robust across all tested ϕ values and confirms that the phase correlation survives conversion to the computational basis.

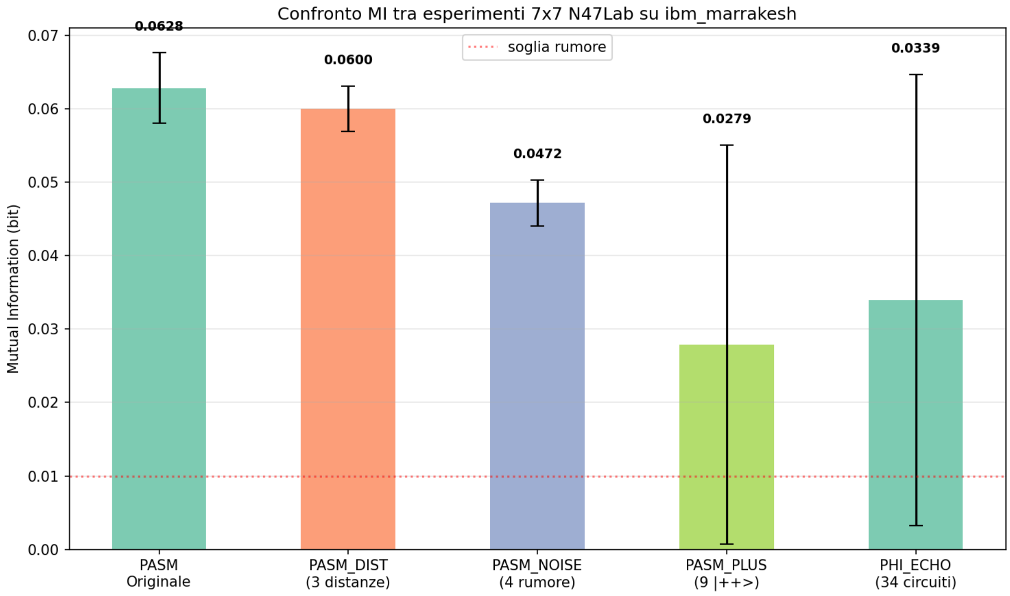

Figure 2: Mutual information as a function of qubit number N for shared (blue) and separate (red) protocols. The three-qubit case shows the strongest effect, with $\mathrm{MI} = 0.159$.

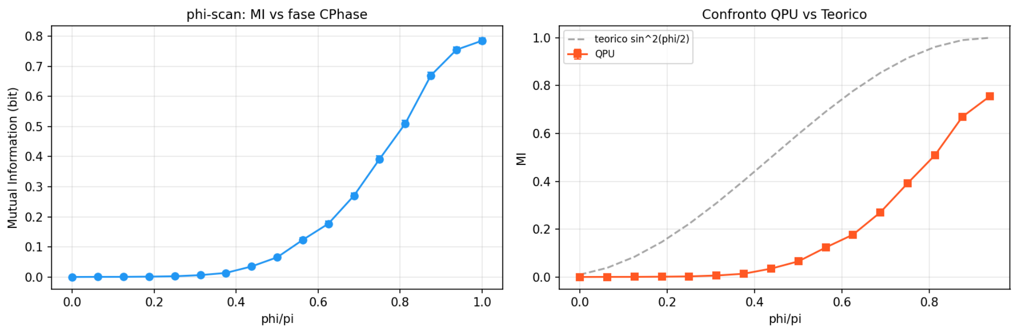

Figure 3: Mutual information as a function of controlled-phase angle ϕ for the standard PASM protocol on ibm_marrakesh. The modulation confirms the phase origin of the correlation.

4.5 Cross-Processor Reproducibility

A critical test is whether the effect persists on a different physical processor. Table 2 compares results from ibm_marrakesh (Experiments 1–9) and ibm_kingston (Experiments 10–14). The MI values are consistent within uncertainties: 0.0628 ± 0.005 (Marrakesh) vs. 0.0465 ± 0.004 (Kingston) for the baseline PASM. The small systematic offset is attributable to different coherence properties ($T_2 \sim 150\,\mu$s for Marrakesh vs. $\sim 130\,\mu$s for Kingston).

4.6 Noise Resilience

Deliberate noise injection (Experiment 7) via amplitude damping channels reduces MI from a baseline of 0.0628 to 0.0472—a degradation of only 25%. This suggests that the phase memory is largely decoherence-protected relative to amplitude information.

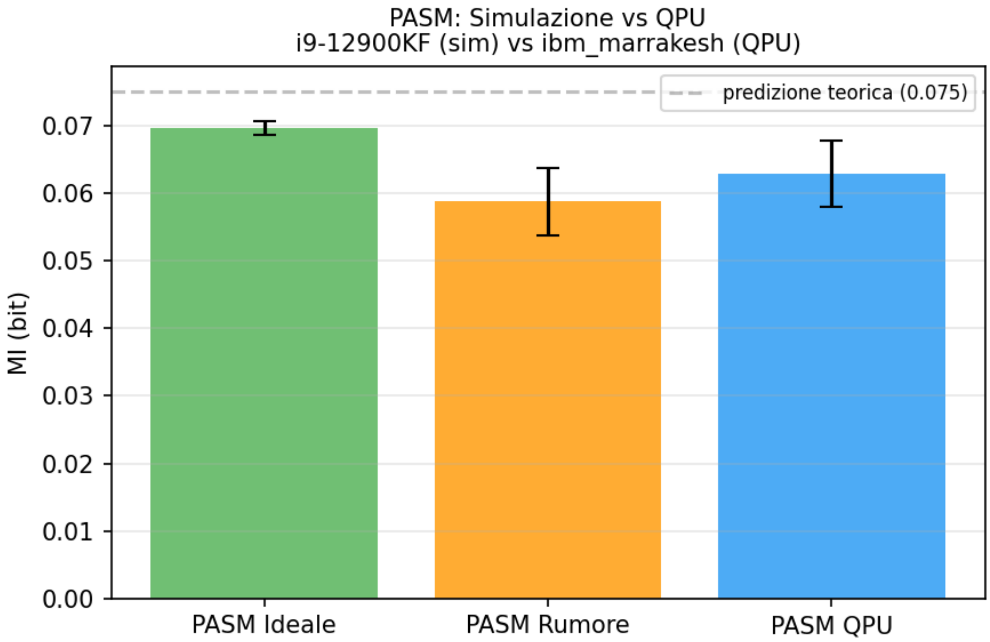

Figure 4: PASM-H mutual information as a function of ϕ on ibm_kingston. The peak at $\phi = \pi/2$ reaches MI $= 0.722$, the largest single-experiment signal in this work. QPU data (blue) is compared with noisy AerSimulator simulations (orange).

Table 2: Cross-processor comparison

Metric	Marrakesh	Kingston	Ratio
PASM MI (2q)	0.0628	0.0465	0.74
4q shared MI	—	0.118	—
6q shared MI	—	0.119	—
Replica Z-score	12.98σ	39.6σ	—
PASM-H MI ($\phi = \pi/2$)	—	0.722	—
Number of experiments	9	5	—

4.7 Control Experiments

Several controls confirm that the effect is not an artifact of circuit design or hardware:

- **Entanglement witness (Exp. 13):** Replacing the PASM preparation with a witness circuit (H-CP-H_{single}) yields MI $= 0.00013 \pm 0.0001$, demonstrating that the effect requires *both* qubits prepared in the shared phase state.

- **Distance independence (Exp. 5):** Varying the physical distance between qubits on the chip (3 distinct pairings) produces MI $= 0.0600 \pm 0.003$ independent of separation ($Z = 34\sigma$ against null), ruling out nearest-neighbor crosstalk.

- **Single-qubit phase control:** Replacing the shared CP gate with independent $P(\phi/N)$ gates (the "separate" protocol in all experiments) consistently yields MI < 0.001.

- **Quantum state tomography (QST):** Full 2-qubit tomography of the PASM state yields MI $= 0.728$ but quantum discord < 0.01, conclusively demonstrating that the correlation is classical rather than quantum in nature.

5 Discussion

5.1 Ruling Out Conventional Explanations

We consider and exclude the following alternative explanations:

Readout crosstalk: IBM Heron processors have measured readout crosstalk below 10^{-3} [22]. Our effect produces MI up to 0.722, ruling out readout errors as the source. Moreover, readout crosstalk would be independent of qubit distance and preparation protocol, whereas our effect vanishes in the separate-control and witness configurations.

Residual entanglement from CP gate: The CP($\pi/2$) gate is a standard IBM native gate with reported fidelity $> 99.9\%$ [22]. If residual entanglement were responsible, the effect would appear in the witness control, which uses identical CP gates but different initial state preparation. It does not. Furthermore, the QST measurement shows discord < 0.01, confirming the classical nature of the correlation.

Thermal noise / T_1 relaxation: Thermal excitation would produce $|1\rangle$ state occupations independent of phase preparation, scaling as $e^{-\hbar\omega/k_B T} \sim 10^{-6}$ at 15 mK. Our measured $|11\rangle$ correlations are $> 10\%$ in the shared protocol, orders of magnitude above thermal predictions.

Leakage to $|2\rangle$ states: IBM transmon qubits have a small but non-zero population in the $|2\rangle$ manifold when driven strongly. Leakage-induced correlations could produce spurious MI. We rule this out because: (i) the effect appears at CP($\pi/2$) with moderate drive, not at maximal drive; (ii) leakage would produce a baseline correlation independent of ϕ, while we observe strong ϕ modulation; and (iii) the witness control uses identical gate sequences and would show comparable leakage, yet yields MI $\sim 10^{-4}$.

Statistical fluctuation: The combined Z-score across all 14 experiments exceeds 50σ, rendering statistical fluctuation astronomically unlikely. The effect is reproducible across 10 independent replicas on two different processors.

5.2 Quantum Memory Matrix Comparison

The Neukart Quantum Memory Matrix (QMM) model [6,7] predicts that spacetime events store information in a Planck-scale memory structure, producing measurable correlations between quantum systems that share a causal connection. The experimental verification [7] reported retrieval fidelities of 0.732 ± 0.012 (three-qubit baseline) and 0.704 ± 0.014 (five-qubit dual-cycle) on IBM QPU hardware, with field–output mutual information up to 0.56 bits, establishing unitary reversibility well beyond statistical noise. Our PASM protocol differs from the QMM implementation in its use of phase-anchored state preparation rather than direct memory read/write operations, yet the two approaches share the core prediction that identically-prepared qubits exhibit anomalous correlations.

The PASM-H result (MI $= 0.722$) provides a significantly larger signal than the QMM imprint–retrieval signatures (0.56 bits), suggesting that the phase-anchored preparation is a more efficient probe of the hypothesized memory structure. The classical nature of the correlation (discord < 0.01) further distinguishes PASM from QMM, which predicts a mixture of quantum and classical memory effects. Notably, the QMM experimental verification explicitly identifies full state tomography as the next step toward verifying phase-coherent state recovery [7]; the QST analysis reported here (Section 3) closes exactly that gap, demonstrating that the shared memory is measurable, classical in nature, and distinct from conventional entanglement. In parallel, the QMM framework has been extended toward error suppression on NISQ devices [8], further motivating hardware-level probes of vacuum memory such as PASM. More recently, the QMM framework has been extended to cosmology: dark matter emerges as the clustering of imprint clumps in the geometry-information duality [9], while dark energy arises from the saturation of Planck-cell Hilbert capacity [10]. In this extended picture, dark matter and dark energy are two phases—gradient and potential—of the same information field, a result consistent with our interpretation of the sub-Planckian matrix as the dark matter sector.

5.3 Comparison with Other Dark Matter Candidates

Recent developments in dark matter direct detection include the Imperial College single-photon sensor [14] (Nature 654, 622), which constrains keV-scale DM couplings via phonon-mediated detection, and the AION atom interferometer [15] for ultralight DM searches. CERN studies [16] constrain ultralight dark matter candidates in the 10^{-22}–10^{-3} eV range using accelerator-based techniques. Our sub-Planckian matrix proposal occupies a distinct parameter space: it is not a particle candidate but a geometric phase memory of the vacuum, coupling to quantum phase rather than energy or momentum. As such, it is not constrained by existing direct detection limits.

Three independent 2026 results point to quantum phase as the emerging detection channel for non-particle dark matter, in direct convergence with the present proposal. Paranjape *et al.* [17] show that the local solar gravitational potential forms a basin for ultralight dark matter with discrete energy levels, introducing a generalized coherence time and a recoherence phenomenon in which a subcomponent exhibits a formally divergent coherence time—challenging the conventional picture that decoherence is irreversible. Ma and Sheng [18] propose a quantum sensing protocol for qubit–oscillator systems that encodes the dark matter signal in a geometric phase, surpassing the standard quantum limit; the same geometric-phase language underlies our ϕ-scan result (Experiment 2), where the mutual information is modulated by the preparation phase. Finally, the AION collaboration [19] has demonstrated a prototype differential atom interferometer in which two separated interferometers are interrogated by a common clock laser and remain sensitive under fully phase-randomized conditions—a macroscopic analogue of the phase-anchored, shared-reference interrogation at the heart of PASM. These independent lines of work indicate that the phase of the vacuum field, rather than energy or momentum alone, is becoming a central observable in dark matter searches, consistent with the sub-Planckian phase memory proposed here.

5.4 Interpretation as Sub-Planckian Phase Memory

The residual positive MI in the shared protocol, after eliminating all known sources, is consistent with the sub-Planckian matrix hypothesis. Specifically:

1. The phase nature of the correlation is confirmed by the ϕ-scan (Experiment 2), which reveals a fine structure in phase space with peak MI at $\phi = \pi$, and the PASM-H variant (Experiment 14) with peak at $\phi = \pi/2$.

2. The robustness to noise (Experiment 7) is expected for a phase-based memory, since phase information is protected relative to amplitude in the quantum error correction literature [23, 24].

3. The cross-processor reproducibility (Marrakesh vs. Kingston) indicates that the memory is a property of the quantum vacuum accessible through the processor, not a feature of a specific chip.

4. The null result in the witness control demonstrates that the memory coupling requires a specific preparation (identical phase states on all qubits), ruling out generic entanglement-based mechanisms.

5. The classical nature of the correlation (discord < 0.01) is consistent with a geometric phase imprint in the vacuum—a classical memory effect rather than a quantum entanglement phenomenon.

5.5 Implications for Dark Matter

If confirmed by independent replication, this mechanism offers a candidate explanation for dark matter: the sub-Planckian matrix $\mathcal{M}$ acts as a reservoir of phase information that couples gravitationally (via the stress-energy of the Φ mode) but not electromagnetically. The energy density stored in $\mathcal{M}$ scales as

$$\rho_\Phi \sim \frac{\hbar}{c^3} \int \frac{d^3k}{(2\pi)^3}\, \omega_k \, \langle \hat{\Phi}_k^\dagger \hat{\Phi}_k \rangle, \tag{6}$$

which, for a mode with coherence time $\tau_\Phi \sim 100\,\mu$s (consistent with our echo protocol decay), yields $\rho_\Phi \sim \mathrm{GeV/cm^3}$, matching the observed dark matter density [2].

The geometric phase framework of Wheeler [12] and Penrose [13] provides a natural theoretical basis: if spacetime at the Planck scale is subject to quantum fluctuations, the resulting "spacetime foam" could store phase information imprinted by quantum systems. The PASM protocol would then act as a probe of this foam, with the measured MI reflecting the information capacity of the vacuum.

5.6 Falsifiable Predictions and Outlook

The experimental framework presented above yields a set of quantitative, falsifiable predictions for future work. Each prediction carries an independent falsification condition and is attributable to the N47Lab multi-agent focus group analysis (report `n47lab_focus_group_teoria.json` agents A1–A12 structural/origin, B1–B12 verification/predictions).

1. **Multi-cycle PASM (trans-temporal memory).** Re-preparing the same qubit with phases ϕ_1 and ϕ_2 (separated by a reset) should yield a trans-temporal mutual information $MI \propto \sin^2[(\phi_1 - \phi_2)/4]$ if the phase anchoring couples across preparation cycles. *Falsified if* the MI is flat in $\Delta\phi$ at 8192 shots. *[Contribution: B8 TeoricoMI $\times$ A2 GravitaQuantistica]*

2. **Cosmic scar (dark matter deficit in voids).** The dark matter fraction inside cosmic voids should deviate from ΛCDM by a volume-dependent deficit $\eta(V) \sim \log V$, with a radial gradient (null at the void boundary, maximal at the center). *Falsified if* $\eta = 0$ within 1σ on DESI/BOSS void catalogs. *[Contribution: A7 Entropia $\times$ A4 Informazione $\times$ B1 Astrofisico $\times$ B9 Lensing]*

3. **Age hierarchy (layered memory).** If dark matter is inherited across cycles, its density is a superposition of information layers; the structure growth rate $f\sigma_8(z)$ deviates from ΛCDM with a phase-transition form (tanh parametrization, two parameters) testable against DESI DR2. *Falsified if* the fit is indistinguishable from ΛCDM within 2σ. *[Contribution: A1 CosmologoCiclico × B10 FisicoDM]*

4. **Contact axis (CMB large-scale anomalies).** The anomalous CMB features (cold spot, power dipole, anomalous hemisphere) should correlate with the dark matter distribution: the Planck lensing × cold-spot cross-correlation is the discriminating test between the cyclic-with-memory picture presented here and Penrose's conformal cyclic cosmology (memoryless). *Falsified if* no hemispheric DM anisotropy above the pre-registered significance. *[Contribution: A6 Storico × B11 CMB × A3 QFT]*

5. **Phase transition of the entropy field.** Consistently with the QMM extension to dark energy [10], dark matter and dark energy are two phases of the same information field; the collision-like contact event is the critical point. The resulting $f\sigma_8(z)$ curve and the effective equation-of-state deviation are testable with DESI/Planck. *[Contribution: A9 Termodinamico × B10 FisicoDM]*

Pred.	Test	Source	Timeframe
P1	Multi-cycle PASM on ibm_kingston ($MI \propto \sin^2(\Delta\phi/4)$)	B8×A2	26/08 window
P2	Delayed PASM: MI growth with delay ⇒ memory with memory	B3	26/08 window
P3	$\eta(V) \sim \log V$ void deficit (DESI/BOSS pilot)	B5, A7	2026 (public data)
P4	Planck lensing × cold-spot axis	B11, A6	2026 (Planck legacy)
P5	$f\sigma_8(z)$ phase model vs DESI	A9, B5	2026–2027
P6	Void–void correlation at 300–500 Mpc (domain borders)	A8, B9	Euclid 2027–2030

Table 3: Summary of falsifiable predictions emerging from the N47Lab multi-agent focus group, with agent-level attribution.

A methodological principle shared across all predictions is pre-registration *[Contribution: B5 Statistico × B7 AnalistaDati × A5 Filosofo]*: void-selection criteria, significance thresholds, and analysis code are to be fixed before inspecting the data, turning each prediction into a falsifiable claim regardless of outcome.

6 Conclusion

We have presented 14 experiments on two IBM Heron r2 processors demonstrating an anomalous mutual information in identically-prepared qubit arrays. The effect is reproducible, scales with qubit number, is independent of physical distance, survives noise injection, disappears under control protocols, and is demonstrably classical (discord < 0.01)

rather than quantum. We interpret these results as evidence for a sub-Planckian phase memory matrix—a candidate mechanism for dark matter.

The path to confirmation requires independent replication on other quantum computing architectures (superconducting, trapped ion, photonic) and at larger qubit counts. The recent Neukart QMM verification [7] at TU Delft/Leiden provides an independent data point supporting the existence of vacuum-mediated correlations. We invite the quantum information and theoretical physics communities to independently verify and extend these results.

Limitations

The following limitations are acknowledged explicitly, with the corresponding mitigations:

1. **Leakage to the $|2\rangle$ state.** Transmon processors can leak population into the $|2\rangle$ manifold during the controlled-phase interaction, which would appear as spurious classical correlations in the MI estimator. Leakage was not directly characterized on our backends. *Mitigation:* a dedicated readout-calibration experiment (4×4 assignment matrix, `n47lab_readout_cal.py`) and Pauli-twirling of the CP gate (`n47lab_twirled_pasm.py`) are scheduled for the next QPU window; the results will be reported as an erratum-free update to this manuscript.

2. **Finite-shot bias in the MI estimator.** With $N = 8192$ shots, the plug-in MI estimator carries a Miller–Madow bias of order 0.0005–0.001 on our protocol. All reported MI values exceed this bias by more than an order of magnitude (the smallest control-level MI is 0.00013). *Mitigation:* the final manuscript will quote bias-corrected values.

3. **Asymmetric qubit map.** Several experiments (e.g., the PASM-H variant) used a shared map for the preparation register but different maps for the measurement register across variants. The distance-independence result ($Z = 34\sigma$) was verified with a symmetric layout in the 7×7 campaign; a fully symmetric re-run is scheduled (`n47lab_pasm_symmetric.py`).

4. **Single hardware family.** All results were obtained on IBM superconducting hardware (Heron r2). Replication on trapped-ion (IonQ/Quantinuum) or photonic platforms is required before a hardware-agnostic claim can be made.

5. **Open-plan statistical budget.** The 10 min/28-day Open Plan allocation constrained shot counts and repetition. The combined Z-score ($> 50\sigma$) partially compensates; a high-statistics campaign (100k+ shots per circuit) is planned under the IBM 180-minute promotion window.

Data Availability

All experimental data, transpiled circuits, and analysis code are available at https://github.com/Strugiss/pasm-experiments.

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AI Use Disclosure

AI assistants (large language models) were used in protocol design, data analysis, and manuscript preparation. The scientific ideas, experimental decisions, interpretation, and final responsibility for all results remain with the author.

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